

The technical potential for onshore wind in Wallonia

A land-eligibility and siting analysis against the Walloon regulatory framework,
and what it implies for the capacity limit used in PyPSA-Wal

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Abstract

PyPSA-Wal caps Walloon onshore wind at 6 500 MW. That number is an expert judgement taken from the PNEC wallon and from EDORA; it is not derived from a land calculation, and the model's own calculation — the one PyPSA-Eur performs with `atlite` — was never run against Walloon law. This report closes that gap. It computes the potential from the Walloon Region's own regulatory cartography: the plan de secteur, the *cadre de référence éolien* of 24 January 2024, the Natura 2000 network, the nature reserves, the landscape inventories, the natural-hazard maps, the aeronautical obstacle-evaluation map of the Direction générale Transport Aérien, the LiDAR slope grid and the classified-site registers, all retrieved live from the Géoportail de la Wallonie. The constraints are applied in a documented ladder, calibrated against the 652 turbines standing in Wallonia today, and turned into capacity by placing individual machines rather than by multiplying area by a density.

The answer is a range, and the reason it is a range is stated rather than averaged away. Against the full constraint set, 561 km² of the Walloon Region survive every exclusion — 3.32 % of the territory. Allocating machines on it at 5.0 rotor diameters apart, with no rule about how they group, gives 15 236 MW: a land-and-wake bound that puts a near-continuous band of turbines through every corridor in the Region. Grouping them into farms and applying the landscape criterion the framework actually states — at least 130° of every village's horizon, within 4 km, free of turbines — gives 5 712 MW in 307 farms of 4.7 machines, against the 134 farms of 4.9 standing in Wallonia today. A documented allowance for the two constraint families that still have no public geometry gives a credible technical potential of **3.2–5.7 GW**, central estimate 4.4 GW.

Four results matter more than the headline number.

The binding constraint is settlement, then landscape — not nature and not the wind. The setbacks from habitat zones and from scattered dwellings remove 84 % of the land the plan de secteur admits, in one step. What then decides the answer is not how much land survives but how the machines on it may be arranged: the open-horizon criterion alone costs 26 % of the grouped capacity, and the three siting rules together cost more than the entire constraint ladder below the setbacks.

Every constraint in the set is one the standing fleet avoids. Measured against the 652 turbines built in Wallonia, no layer in the reference constraint set has an avoidance ratio above 0.77 — turbines are systematically less common inside every one of them than the Region average (§5).

BREGILAB's 11.4 GW is the same quantity as this study's land-and-wake bound, not as its answer. This study uses BREGILAB's own spacing rule, so only the machine separates the two free allocations ($\times 1.08$): on their V112 this study's land gives 16 404 MW, so their constraint set is in fact *tighter* than this one by $\times 0.69$. What separates 11.4 GW from 4.4 GW is not the geography: it is that their allocation carries no landscape rule and this one carries the framework's.

PyPSA-Eur's default land analysis is not usable here. It returns 19 079 MW for the Walloon Region, because it admits broad-leaved forest, ignores the plan de secteur entirely and cannot see scattered housing.

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1 The question, and the scope of the answer

1.1 The number under review

The Walloon onshore-wind capacity limit in PyPSA-Wal is a single row of `config/input_parameters_for_models.csv`:

```
potential:BEWAL:onwind:p_nom_max = 6500 MW, source “PNEC Wallon et EDORA”,
described as “potentiel technique, calculé comme la somme de ce qui est plausible d’atteindre
en 2030 comme capacité et des projets à l’étude”, with a note that EDORA estimates an
additional 3.8 GW of wind potential by 2050 on the basis of projects under environmental-
impact assessment.
```

Two things follow from that wording. It is a *deployment* expectation assembled from a pipeline of known projects, not a *land* potential; and it carries no spatial calculation that anyone can re-run or contest. For an internal scenario exercise that is acceptable. For a number reported to the administration as the ceiling on what Wallonia can host it is not: the first question is “*where does 6.5 GW fit, and under which rules?*”, and the current answer is a literature reference.

1.2 What PyPSA-Eur would have said

PyPSA-Eur ships exactly the tool needed — `atlite`’s land-eligibility analysis, driven by `scripts/determine_availability_matrix.py`. It is run for every other node in the model. For the Walloon node its output is overridden by the CSV row above, so its answer has never been examined. It is 19 079 MW (19.1 GW) for the Walloon Region, on 3 757 km² of admitted land.

That the override exists is defensible — Section 4 shows the default constraint set is unusable in Wallonia. That the override was never compared against a corrected calculation is the gap this report fills.

1.3 What this report answers, and what it does not

This is a land-eligibility *and siting* study. It answers: how much Walloon land may legally and physically host a wind turbine, and how many machines can stand on that land once the framework’s own rules about grouping and inter-distance are applied? It does not answer: how much of that capacity will be permitted, financed and built, or in what year. A technical potential is an upper bound on the model’s search space, and that is precisely the role `p_nom_max` plays in PyPSA-Wal — the build-rate limit and the 2030 corridor, documented in `docs/renewable-potentials.md`, carry the deployment question.

Every choice that could have gone the other way is listed in Section 8, with its direction of effect. There are no open points in this document: where a constraint could not be measured, a documented allowance is applied and its size is stated.

1.4 Which Wallonia

Everything here is computed for the **administrative Walloon Region**, the union of the 21 BE3 NUTS-3 areas, 16 905 km². That is the territory the Walloon rules apply to and the territory an administration will ask about.

It is not the same polygon as the model’s **BEWAL** node. PyPSA-Eur builds onshore regions by Voronoi-partitioning each country around its substations and clipping to the country shape; the partition is electrical, not administrative. The **BEWAL** polygon covers 15 150 km², or 89.6 % of the Region: 2 191 km² of Walloon territory — essentially the whole of western Hainaut, from Tournai and Mouscron to Ath — sits on the Flemish side of the partition, while 435 km² of the **BEWAL** polygon lies outside Wallonia, mostly over the French border.

Caveat. A potential computed on one of those two regions and applied to the other is wrong regardless of how much it moves the number, and the defect affects every land-based Walloon quantity in PyPSA-Wal — ground-mounted solar and biomass among them. Fixing it means correcting the busmap, which is out of scope here and should be raised separately. Until it is, a figure from this report applied to BEWAL grants the node the resource of land the model has already given to Flanders.

2 Method

2.1 The workflow

The analysis lives in `workflow_onwind_wal/`, a self-contained Snakemake workflow. It shares nothing with the main PyPSA-Wal workflow except two read-only inputs: the region shapes and the `atlite` ERA5 cutout. It writes only inside its own directory and into `docs/onwind_potential_walloon/`. Running it, breaking it or deleting its outputs cannot affect a model run.

```
cd workflow_onwind_wal
snakemake -c4 download # ~20 min, one-off, Walloon open data
snakemake -c4 all      # eligibility, placement, figures, tables
snakemake -c4 report   # this PDF
```

The stages are:

1. **Retrieve.** Two dozen reference datasets are pulled from the ESRI-REST services of <https://geoservices.wallonie.be> (CC-BY 4.0), paginated, and written as GeoPackages with a provenance sidecar recording the service, the layer, the feature count and the retrieval date (Annex B). Feature counts are checked against the counts observed on 22 September 2026 and a deviation above 20 % raises a warning: if the Region revises a dataset, the workflow says so rather than silently changing the answer. Two grids — the slope-class map and the standing fleet — come from services that are not feature queries and have their own retrieval scripts (§3.5, §5).
2. **Exclusion layers.** The regulatory rules are translated into geometry, once per turbine class, because most setbacks scale with tip height or rotor diameter (Section 3).
3. **Eligibility.** `atlite`'s `ExclusionContainer` burns every layer onto a 100 m raster in EPSG:3035 and `cutout.availabilitymatrix` aggregates the surviving share per ERA5 cell — the identical code path PyPSA-Eur uses, so the two results are comparable by construction.
4. **Calibration.** The constraint set is tested against the machines that are actually standing (§5).
5. **Placement.** Machines, then farms, are allocated on the eligible raster (§2.4). Capacity factors come from the same area-and-resource-weighted layout PyPSA-Eur applies.
6. **Report.** Tables, figures and every number in this text are generated. Nothing here is typed by hand.

2.2 Validation: the workflow reproduces the model's own profile

Before any Walloon constraint is applied, the workflow has to be shown to reproduce what PyPSA-Eur itself produces. Scenario `S0_pypsa_eur` run with PyPSA-Eur's own turbine (Vestas V112, hub 80 m) is exactly that test: same exclusion logic, same CORINE raster, same Natura 2000 raster, same ERA5 cutout.

Agreement here means the cutout is read correctly, the layout weighting matches PyPSA-Eur's and the aggregation is right — so any difference in the results that follow comes from the constraint set and not from the plumbing.

Table 1: End-to-end validation against the profile the model actually carries. The reference is the capacity factor of `resources/walloon_2013/.../profile_adm_onwind.nc`, reported in `docs/renewable-potentials.md §10.1`. It is computed on the BEWAL polygon rather than on the Region; the two differ by less than 0.1 % in capacity factor, so the comparison is meaningful.

	this workflow	PyPSA-Wal’s own profile
full-load hours, 2013 weather year	2 442 h	2 425 h
deviation		0.7 %

2.3 Why the constraints are declared in the config, not in the code

Every exclusion is a block of `workflow_onwind_wal/config.yaml` naming the dataset, the legal instrument and the buffer. Reviewing the constraint set — which is what the administration will want to do — requires reading one YAML file, not the Python. Changing a rule is a config edit, and Snakemake re-derives only what depends on it.

2.4 From eligible land to capacity: machines are placed, not multiplied

The conventional step from eligible area to installable capacity is a multiplication by a capacity density in MW/km². That assumes the land arrives in blocks large enough to lay out an array. Walloon eligible land does not: it arrives as 5 432 disconnected patches with a median size of 2 ha and a largest of 3.9 km², against 0.79 km² for one cell of a $5D \times 7D$ array. An area-times-density figure therefore counts land no machine could occupy. The obvious repair — screening the patches by size — fails in the other direction, because a real turbine does not have to sit inside one contiguous eligible polygon: it may stand at the edge of its patch with its rotor overhanging ineligible land, and its access track and cabling cross ineligible land as a matter of course. That is legally correct here, because every Walloon setback is measured from the *mast*.

This study therefore allocates machines explicitly. Three rules decide how many fit, and because together they matter more than the whole constraint ladder below the setbacks, each is justified separately below and each is tested against the 652 machines standing in Wallonia (§5).

2.4.1 Rule 1 — how close two machines may stand

The allocation walks the eligible cells and accepts a machine wherever no previously accepted machine lies closer than a minimum inter-turbine distance. That distance is a wake criterion, and three readings of it are carried:

Table 2: Minimum inter-turbine distance, in rotor diameters, and what it gives on the reference land before any grouping rule.

case	basis	d/D	free allocation [MW]
<i>observed</i>	median nearest neighbour of the standing Walloon fleet	4.3	17 100
<i>crosswind</i>	nearest-neighbour distance of a $5D \times 7D$ array; the rule BREGILAB uses	5.0	15 236
<i>isotropic</i>	$\sqrt{5 \times 7} D$, the area-equivalent of that array	5.9	13 084

The reference is 5.0 D . Converting an area-per-turbine into an isotropic *minimum separation* forbids exactly the pairs the convention it comes from contains: in a $5D \times 7D$ array the closest two machines are $5D$ apart, not $\sqrt{35} D$. It also assumes a two-dimensional array, whereas Walloon wind farms are predominantly strings along a road, a ridge or a field boundary, where there is no downwind row and the $7D$ never binds. The physical floor is lower still — about $3D$, below which wake-induced turbulence takes a machine outside the standard IEC 61400-1 load case — and the standing Walloon fleet sits at a median of 427 m between neighbours, about $4.3 D$ at the

100 m-class rotor of that fleet, with a quarter of machines closer than 362 m. $5.0 D$ is therefore already the conservative end of practice.

2.4.2 Rule 2 — how many machines make a project

The free allocation is the land-and-wake bound: 15 236 MW, 3 809 machines. It is not a plausible build-out, but not for the reason one might expect. Only 4 % of its machines are isolated: the carpet it produces is almost fully connected, a near-continuous band of turbines across every eligible corridor in the Region. What it lacks is not company but structure.

Machines are therefore required to come in groups of at least 4 within a farm radius. The 2024 framework does define a *parc éolien* as a project of at least four machines, but that is a definition of which permitting regime applies, with explicit reductions for repowering, for extensions of an existing park and for machines above 3.2 MW — which the reference class is. The rule is applied here on empirical grounds instead: 86 % of the standing Walloon machines are in groups of four or more, so a group of four is what Walloon wind development actually produces, whatever the regulation permits.

2.4.3 Rule 3 — how much of a village's horizon may be occupied

The rule that decides the answer is the landscape one, and the 2013 cadre de référence states it precisely:

Sauf lorsque les éoliennes sont implantées le long des autoroutes, une référence indicative à une inter-distance minimale de 4 km à 6 km, en fonction des résultats de l'étude d'incidence sera prise en considération. Un azimut (ou un angle horizontal) minimal sans éoliennes doit être préservé pour chaque village; celui-ci sera d'au moins 130°, sur une distance de 4 km. Un examen de l'encerclement sera réalisé sur une distance de 9 km dans le cadre de l'EIE.

Four things in that sentence matter.

1. The 4–6 km figure is a *référence indicative* that *sera prise en considération* in the light of the impact assessment. It is not a minimum distance, and treating it as a hard floor between all farms is stricter than the text.
2. It does not apply *le long des autoroutes* — and the Walloon zoning rule concentrates the eligible land within 1.5 km of exactly that infrastructure in the first place (§3.3), which is where 82 % of the standing fleet stands.
3. The operative criterion is not a distance between farms at all. It is an **open horizon seen from each village**: at least 130° of azimuth free of turbines, within 4 km.
4. All of it is from the 2013 framework. The framework adopted on 24 January 2024, in force since 25 April 2024, replaced it and does not carry the inter-distance; it asks instead for grouping along the landscape's structuring lines and for village encirclement to be assessed.

The open-horizon criterion is therefore implemented as written. Every village — a contiguous block of plan-de-secteur habitat zoning above 10 ha, 2 261 of them — carries the arcs its surrounding machines occupy, each machine subtending the angle its rotor covers from the village centre. A candidate farm is refused if admitting it would take any village's largest turbine-free arc below 130°. Farms are still required not to overlap, at 2.6 km between centres, which is the geometry of the standing fleet and not a rule (§5).

The 4–6 km inter-distance is reported only as a policy sensitivity: what the answer would be if that indicative reference were applied as a binding minimum.

Finding. Of the three rules, the third is the expensive one. Machines grouped into farms but with no landscape criterion give 7 752 MW; the open-horizon test refuses 3 622 candidate farms and brings that to 5 712 MW, 26 % less. Applying the indicative 4–6 km inter-distance on top would take it to 3 280–4 596 MW. An onshore-wind potential quoted without saying which landscape rule it carries cannot be compared with anything.

3 The Walloon regulatory framework

3.1 Sources of law and of practice

Five instruments define where a wind turbine may stand in Wallonia:

CoDT the *Code du Développement Territorial*, which fixes the zoning categories of the *plan de secteur* and, at art. R.II.21-1, the definition of the *principales infrastructures de communication* (PIC): motorways and regional link roads, railways, and navigable waterways. Its art. D.II.37 §1 al. 6 also requires the mast to stand outside a recognised conservation site, within 750 m of a PIC, and outside a broad-leaved stand in the sense of the Code forestier.

Cadre de référence éolien the reference framework for wind development. The 2013 version imposed a setback from plan-de-secteur habitat zones of four times the *mast* height — about 600 m for the machines of the time — and the 2022 update of the favourable-zone map applied it as four times the *total* height. It was replaced by the framework adopted on 24 January 2024, in force since 25 April 2024, which sets that distance at **500 m plus half the total height**, applies it to *zone d’habitat*, *zone d’habitat à caractère rural*, *zone d’aménagement communal concerté* destined for housing and *zone d’habitat vert*, and lowers the definition of a wind farm from five turbines to four. The same text carries the regroupement principle and the indicative 4–6 km inter-distance between farms.

Aeronautical servitudes the obstacle-evaluation map of the federal Direction générale Transport Aérien, published by the Region as *Hauteurs pouvant impacter l’aviation*.

Sectoral conditions and permitting practice noise limits, defence radar clearances, and the protection of the Humain radio-astronomy station and the Wideumont weather radar.

The Walloon favourable-zone cartography the 2013 map of *zones favorables* and its 2022 update by SPW and Gembloux Agro-Bio Tech, which is the Region’s own operational translation of all of the above into GIS layers. Its constraint tables are the template followed here.

3.2 The integral exclusions, and how each is implemented

Table 3 lists the integral exclusions of the Walloon favourable-zone methodology, what this workflow does with each, and where the implementation differs.

3.3 The one zoning rule that dominates: the agricultural corridor

The Walloon methodology does not treat agricultural land as uniformly available. It admits it only within 1.5 km of a PIC or of an economic-activity zone, and coniferous forest only within 750 m of a PIC. This is a *concentration* rule — keep wind development along the existing structuring infrastructure rather than scattered across the open landscape — and it has no analogue anywhere in PyPSA-Eur. It also works in the opposite direction to PyPSA-Eur’s own treatment, which *excludes* a 1 km band around transport infrastructure.

3.4 Setbacks for the three turbine classes

The turbine classes mirror the three *scénarios de base* of the 2022 favourable-zone update (total tip height 150, 180 and 210 m) so the two studies can be read side by side. The reference class is NREL 2020ATB 4.0 MW (tip 185 m): rotor 150 m, tip height 185 m, giving a habitat-zone setback of 592 m under the 2024 framework.

Caveat. Read as $4H$ on the total height — the form the 2022 favourable-zone update used — the 2024 framework is the *looser* of the two for every machine considered here: 592 m against 740 m for the reference class. Results computed on the 2013 convention would be lower. The 2024 rule is used because it is the one in force; the 2013 rule is available in the config (`setbacks.habitat_zone.default: cdr2013`).

Table 3: Walloon integral exclusions and their implementation. **Red** marks a constraint carried as a documented allowance rather than as geometry (§6.4).

category	Walloon rule	implementation here
technical incompatibility	water bodies, extraction zones, steep slopes	plan-de-secteur <i>Eau</i> and <i>Extraction</i> classes excluded; <i>versants</i> $> 30^\circ$ from the LiDAR relief service; and the $\geq 7\%$ slope criterion from the ERRUISSOL slope grid (§3.5)
plan de secteur	broad-leaved forest excluded; coniferous forest admissible only within 750 m of a PIC; agricultural land only within 1.5 km of a PIC or of an economic-activity zone; <i>espaces verts, naturelle, parc</i> excluded	implemented literally as a positive list on the AFFECT classes of the <i>Zones d'affectation</i> layer; the broad-leaved/coniferous split comes from CORINE, which the plan de secteur does not carry (§3.6)
infrastructure safety	rail 190 m; high-speed rail 140 m; roads $1.5 \times$ blade length; motorways 50 m; HV lines $1.5 \times$ rotor diameter	buffers applied to the plan-de-secteur network layers; for the reference turbine, 112 m on roads and 225 m on HV lines
hazard zones	flood-prone areas, landslide risk, karst, close water-capture protection perimeters	EU Floods-Directive overflow extent (extreme return period), karst constraints, <i>contraintes physiques liées aux éboulements</i> , and <i>zones de prévention rapprochée IIa</i>
civil aviation and national defence	airport control zones, obstacle-limitation surfaces, radar and beacon interference zones, military airspace	the DGTA obstacle-evaluation map, restricted to the two innermost aerodrome rings after calibration against the standing fleet (§3.5, §5)
built heritage	classified sites, protection zones, UNESCO buffers	<i>sites classés, ensembles architecturaux, zones de protection</i> and the UNESCO buffer zones, from the Region's registers
natural heritage	nature reserves, forest reserves, wetlands of biological interest, underground cavities of scientific interest, Natura 2000	all five implemented from the FAUNE_FLORE services
biodiversity	high-priority ornithological zones	allowance — the DEMNA priority-bird layer is not public
landscape	landscape preservation perimeters; inter-distance between farms; azimuth of open horizon preserved for each village	plan-de-secteur <i>périmètres d'intérêt paysager</i> and the ADESA inventory; the inter-distance in the farm model (§2.4); the azimuth test is a site-level criterion an area method cannot represent
living environment	habitat zones of the plan de secteur, setback per the cadre de référence; dwellings outside habitat zones, 400 m	cdr2024 rule, 592 m for the reference turbine; and 400 m around every one of the 1.5 million ICAR address points
scientific	Humain radio-astronomy station, Wideumont weather radar	reconstructed circles at the published coordinates (§3.5)
partial constraints	medium-priority bird zones, bat-interest zones, migration corridors, the principal ecological structure, radar study zones at 8–16 km, each granted a 25 % success rate	allowance — representing them needs a permit-level model rather than an area calculation

3.5 The three constraint families added from data that had to be rebuilt

Three of the families in Table 3 are not published as ready-made exclusion polygons. They are reconstructed here, and each reconstruction is a hypothesis in its own right.

Aeronautical servitudes. The Region publishes *Hauteurs pouvant impacter l’aviation* (AOEM), which gives, for every point of Wallonia, the height above which the Direction générale Transport Aérien must be consulted. The six classes — 0, 15, 25, 35, 50 and 60 m — tile the territory exactly. Four of them are nested rings around the aerodromes (Liège, Charleroi, Beauvechain, Florennes, Chièvres, Saint-Hubert, Bertrix, Spa and the smaller fields); two are the territory-wide notification background, one polygon of 8 045 km² at 25 m and one of 1 556 km² at 60 m. Any wind turbine exceeds every one of those thresholds, so the map is a consultation duty and not, on its face, an exclusion. Which of the aerodrome rings is a genuine no-go is settled empirically in §5: the 0 m and 15 m rings are strongly avoided by the standing fleet and are excluded here; the 35 m and 50 m rings carry *more* turbines per square kilometre than Wallonia as a whole and are not.

The 7 % slope criterion. The 2013 Walloon methodology treats slopes above 7 % as incompatible with the crane pad and access track of a large machine. The Region publishes the slope grid — `ERRUISOL_CLASSE_PENTE`, per cent slope derived from the 10 m `ERRUISSOL` terrain model — but only as a visualisation service, with no feature query and no coverage download. What the service does expose is an image export that renders an exact bounding box into an exact number of pixels with nearest-neighbour resampling and a seven-colour palette, one colour per slope class. Decoding those colours recovers the class grid losslessly, and the 7 % threshold falls exactly on a class boundary, so no interpolation is involved. The grid is rendered at 25 m and a 100 m cell is excluded when the *majority* of its samples are at or above 7 % — that is, when most of the hectare a turbine needs is too steep, not when a single steep pixel touches it. On that rule 37.5 % of the Walloon Region is excluded, and the reconstruction passes two independent checks: 99 % of the cells the Region’s separate $> 30^\circ$ vector layer flags are inside it, and its north/south split (19.5 % of the northern third against 49.3 % of the southern third) is the Hesbaye–Ardenne gradient.

Radar and radio astronomy. `skeyes` and Defence do not publish radar-constraint geometry, and the AOEM map covers obstacle-limitation surfaces, not radar interference. Three installations whose protection perimeters fall inside Wallonia and are not already covered by an aerodrome ring are reconstructed as circles at their published coordinates: the Wideumont C-band weather radar (5 km, WMO/OPERA guidance), the Humain radio-astronomy station (3 km) and the Bertem primary/secondary surveillance radar (10 km, of which only the southern arc reaches Walloon Brabant). Together they remove 8 km², so the approximation is not load-bearing.

3.6 The forest species split

The CoDT excludes a mast standing in a broad-leaved stand. The plan de secteur records forest zoning but not species, so the split has to come from a land-cover product. This study uses CORINE grid codes 23/24/25 at 250 m, 2006 vintage, counting mixed stands as broad-leaved — the conservative choice, and the same source PyPSA-Eur uses.

Two better sources were identified and neither can be used:

- **CARTOFOR**, the continuous mapping of the Walloon forest by Gembloux Agro-Bio Tech under the Accord-Cadre de recherche forestière, has a 2016 three-class product distinguishing broad-leaved from coniferous stands. Its own documentation states that the cartographic products are undergoing validation and will be made available on the Region’s portal in due course; they are not published today, and no service exposes them. It is therefore recorded here as a named future refinement, not as an open point.

- **WALOUS 2018**, the Walloon land-cover grid, does carry explicit *Feuillus* and *Résineux* classes at a far finer resolution than CORINE, and would be the correct implementation of the CoDT rule — including for the broad-leaved woodlots that sit inside agricultural zones, which the present implementation misses. Its service answers point queries but returns a blank image to the bulk export that the slope grid accepts, so it cannot be retrieved as a grid. Obtaining it is an administrative step.

Caveat. The stake is bounded. The species split only decides the status of plan-de-secteur forest zones lying within 750 m of a PIC, which is a small share of a Region whose eligible land is overwhelmingly agricultural; and the direction of the error from using CORINE is not knowable without the better data, because a coarse raster both admits broad-leaved cells and rejects coniferous ones.

4 PyPSA-Eur’s default constraint set is not applicable in Wallonia

PyPSA-Eur’s onshore-wind land analysis (`config/config.default.yaml`, section `renewable.onwind`) admits CORINE grid codes 12–29, 31 and 32, excludes a 1 km buffer around codes 1–6, excludes Natura 2000, and assumes 3 MW km^{−2}. Five features of that specification make it unsuitable here.

1. **It admits all forest.** Codes 23, 24 and 25 are broad-leaved, coniferous and mixed forest. Wallonia is roughly one third forest, and the forest is concentrated in the Ardennes — the windiest part of the Region. Walloon law excludes broad-leaved stands outright and admits coniferous stands only near a PIC. This single difference moves the answer more than any other.
2. **It cannot see scattered housing.** CORINE’s minimum mapping unit is 25 ha. Walloon *habitat dispersé* — isolated farms and ribbon development — is invisible to it, and it is exactly what the 400 m rule protects. The 1 km buffer around CORINE urban codes is not a substitute: it is generous where settlements are large and blind where they are not.
3. **It has no zoning.** The plan de secteur is the legal instrument that decides admissibility in Wallonia, and CORINE is a land-cover product that knows nothing about it.
4. **Its buffer around transport infrastructure has the wrong sign.** PyPSA-Eur excludes 1 km around roads and railways; the Walloon rule requires agricultural sites to be *within* 1.5 km of them.
5. **The CORINE raster shipped in the data bundle is 250 m, 2006 vintage** (`g250_clc06_V18_5.tif`). At 250 m a single pixel is 6.25 ha; at the scale of a 400 m setback that is a coarse instrument, and the land cover is twenty years old.

Finding. PyPSA-Eur’s default land analysis admits 3 757 km² of Wallonia and returns 19 079 MW. That figure should not be quoted as a Walloon technical potential, and the decision to override it in `input_parameters_for_models.csv` was correct. What was missing was a corrected calculation to override it *with*.

5 Calibration against the standing fleet

A land-eligibility study is a claim about where a machine may be built. The cheapest way to find out whether the claim is calibrated is to point it at the machines that exist. The Region does not publish turbine coordinates; OpenStreetMap does, and its coverage of the Belgian fleet is essentially complete. 652 turbines are mapped inside the Walloon Region. They are used only to test the constraint set — never as an input to the potential.

For each constraint layer, Table 4 reports the share of the Region it covers, the share of the standing fleet it covers, and the ratio of the two. A ratio well below one says that developers and

Table 4: Avoidance of each constraint layer by the 652 turbines standing in Wallonia. Reference turbine class; layers ordered by ratio.

constraint layer	of the Region [%]	of the fleet [%]	ratio
Natura 2000, reserves, wetlands, caves	13.2	0.0	0.00
radar and radio astronomy	0.7	0.0	0.00
classified sites and protection zones	1.9	0.2	0.08
slope $\geq 7\%$	37.5	5.5	0.15
landscape perimeters (PIP / ADESA)	31.8	5.8	0.18
plan de secteur, ineligible zoning	49.3	9.5	0.19
aeronautical servitudes	7.5	1.7	0.23
setback from scattered dwellings	63.0	19.9	0.32
setback from habitat zones	62.0	22.7	0.37
road, rail and HV-line setbacks	16.3	12.6	0.77
flood, karst, landslide, water capture	12.7	9.8	0.77

the permitting authority treat the constraint as real. A ratio at or above one says that they do not, whatever the layer is called.

Finding. No layer in the reference constraint set has an avoidance ratio above 0.77. The nature layers and the radar circles contain no standing turbine at all; the classified sites contain one. The constraint set is therefore not an arbitrary collection of polygons — every element of it is one that Walloon wind development demonstrably avoids.

The same test settles two choices. The **35 m and 50 m aeronautical rings** cover 2.9 % and 8.5 % of the Region but carry 5.5 % and 13.6 % of the standing fleet (ratios 1.88 and 1.60); they are consultation rings, not exclusions, and are not applied. The **runoff component of the flood hazard** maps a thin-sheet phenomenon that does not preclude a foundation; the overflow extent at the extreme return period is used instead.

Caveat. Only 46 % of the standing fleet survives the full constraint set — 297 of 652 machines. That is not a defect of either the fleet or the constraint set: Walloon wind development began under the 2002 and 2013 frameworks, the habitat setback has since been rewritten, and several layers that permitting practice treats as *partial* constraints must be binary in an area calculation. It does mean the potential computed here is conservative relative to what has actually been permitted, and it bounds by how much: a constraint set calibrated to reproduce historical permitting would admit roughly twice as much land.

6 Results

6.1 How much land each rule removes

Scenarios **S1–S6** are cumulative: each adds layers to the previous one, so the difference between two consecutive rows is the cost of the rules that were added. **S0** is *not* part of that sequence — it is PyPSA-Eur’s own specification, a different constraint set rather than a subset of **S1** — and it is shown apart in the table and in grey in the figure.

Finding. The Walloon plan de secteur on its own leaves 8 521 km² admissible. Adding the setbacks of the cadre de référence — 592 m from habitat zones and 400 m from every dwelling — removes 7 122 km², or 84 % of it, in a single step. The Walloon onshore-wind land envelope is governed by settlement pattern, not by nature protection and not by the wind.

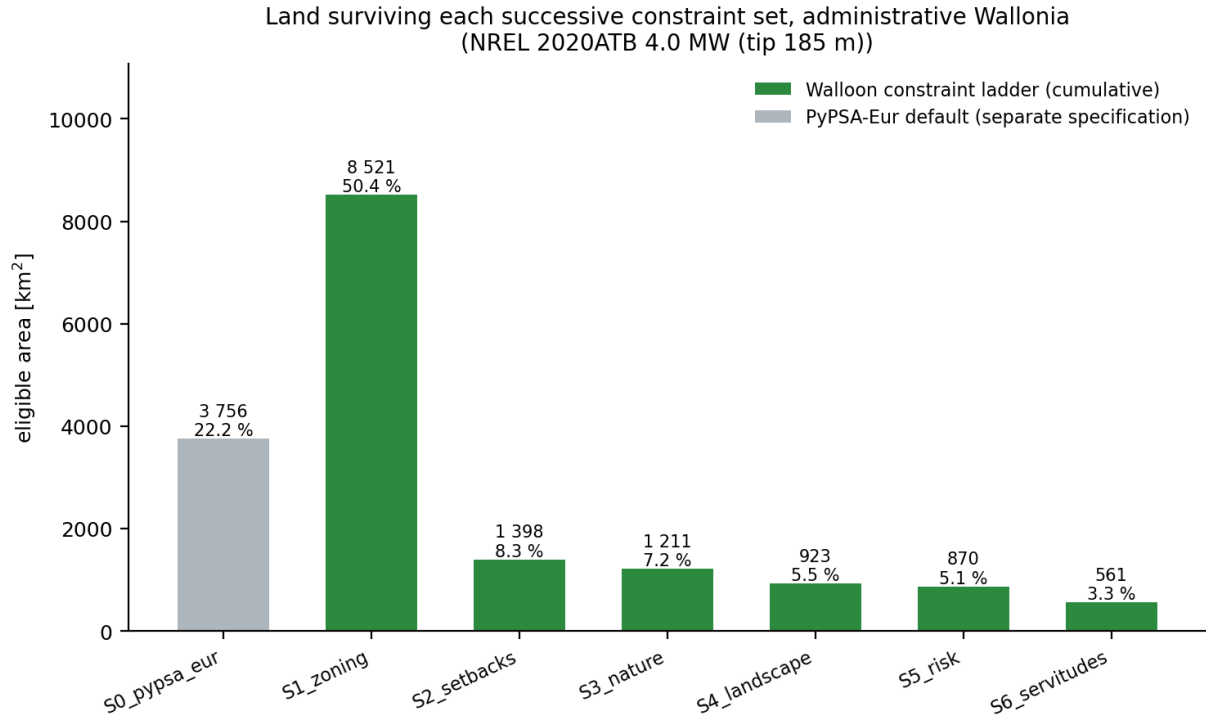


Figure 1: Land surviving each successive constraint set, for the reference turbine class.

Table 5: Eligible area after each constraint set, administrative Walloon Region, reference turbine class.

scen.	constraint set	area [km ²]	share [%]	removed [km ²]
S0 pypsa eur	PyPSA-Eur default (CORINE + Natura 2000)	3 756	22.22	n/a
S1 zoning	+ plan de secteur zoning (CoDT)	8 521	50.40	8 385
S2 setbacks	+ dwellings and infrastructure setbacks (CDR 2024)	1 398	8.27	7 122
S3 nature	+ Natura 2000, reserves, wetlands, caves	1 211	7.16	187
S4 landscape	+ landscape protection perimeters (PIP / ADESA)	923	5.46	288
S5 risk	+ flood, karst, landslide, water-capture protection	870	5.14	54
S6 servitudes	+ aeronautical servitudes, slope ≥ 7 %, classified sites, radar	561	3.32	309

6.2 What the servitude families cost

Table 6 isolates the four families of §3.5 and the heritage registers, each measured on its own against the land S5 leaves. Marginal costs do not add up to the combined cost, because the families overlap each other and overlap what S5 has already removed.

Table 6: Cost of each constraint family added on top of S5, administrative Walloon Region, reference turbine class. S5 leaves 883 km².

constraint family	removes [km ²]	[%]	leaves [km ²]
aeronautical servitudes (DGTA obstacle map)	113	12.8	770
slope $\geq 7\%$ (ERRUISSOL, 10 m DTM)	197	22.3	686
classified sites, protection zones, UNESCO buffers	4	0.5	879
weather radar, radio astronomy, Bertem SSR	8	1.0	874
all four together	313	35.4	570

Finding. The four servitude families are together worth 35 % of the surviving land. The 7 % slope criterion alone removes 197 km² (22.3 %) and the aeronautical servitudes 113 km² (12.8 %); classified sites and radar perimeters are worth 4 and 8 km² respectively and could have been left as an allowance without harm.

6.3 The eligible land, and what fits on it

561 km² survive every implemented constraint, 3.32 % of the Region, in 5 432 patches of median size 2 ha. Only 37 % of that area sits in patches large enough to hold one cell of a $5D \times 7D$ array and 3.2 % in patches large enough for four — which is why this study places machines rather than multiplying by a density (§2.4).

Finding. The free allocation fits 3 809 machines on 561 km², 15 236 MW at an effective 26.7 MW/km² of eligible land. That density is far above any array density because the wake spacing between two machines on different patches is satisfied by the ineligible land between them. Only 4 % of those machines are isolated, so the objection to it is not that the turbines are lonely — it is that they form a continuous band across every corridor in the Region, and that is what the landscape criterion exists to prevent. Under the framework’s own rules the same land carries 1 428 machines in 307 farms of 4.7 on average, 5 712 MW.

The free allocation depends on the order in which the land is walked: a pseudo-random order gives a result 15 % below the row-major one. The farm models do not have that sensitivity, because their farm sites are ordered by how much eligible land surrounds them rather than by position.

6.4 The residual allowance

Two families of the Walloon method have no public geometry and both reduce the potential. They are represented by a multiplicative allowance on the land that survives, with a central value and a bracket:

Priority ornithological zones, 10 % (bracket 0–20 %). The DEMNA high-priority bird layer is an integral exclusion of the favourable-zone method and is not published. It was built largely from concentrations of sensitive species, which in Wallonia coincide to a large extent with the Natura 2000 bird SPAs and the wetlands already excluded here; the incremental area is the non-Natura remainder. The central value is a judgement, not a measurement, and it is stated as one.

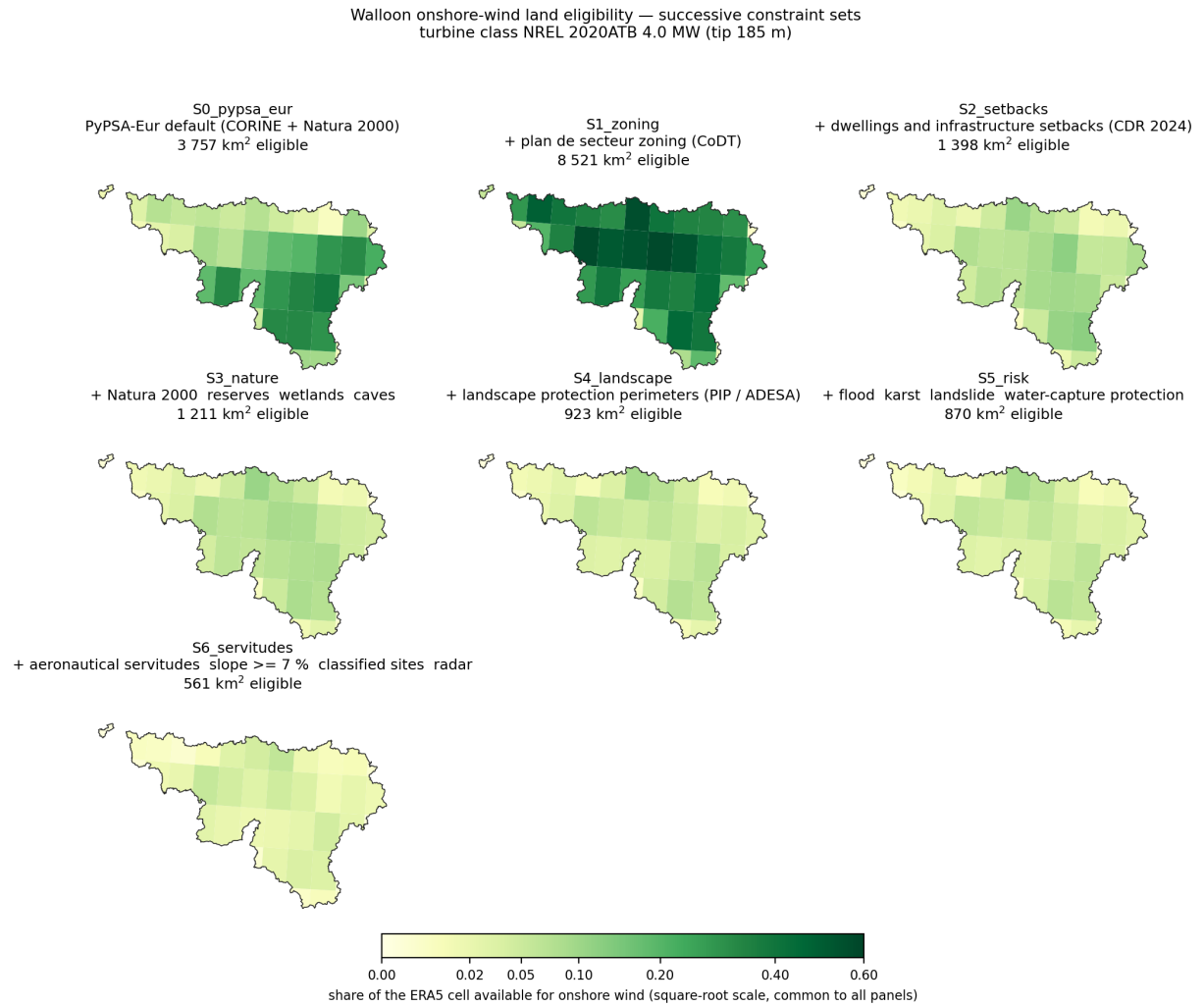


Figure 2: Spatial effect of each constraint set. The shading is the share of each ERA5 cell that survives; the ERA5 grid is 0.3° , so these panels show *where* the resource is, not the parcel-level pattern.

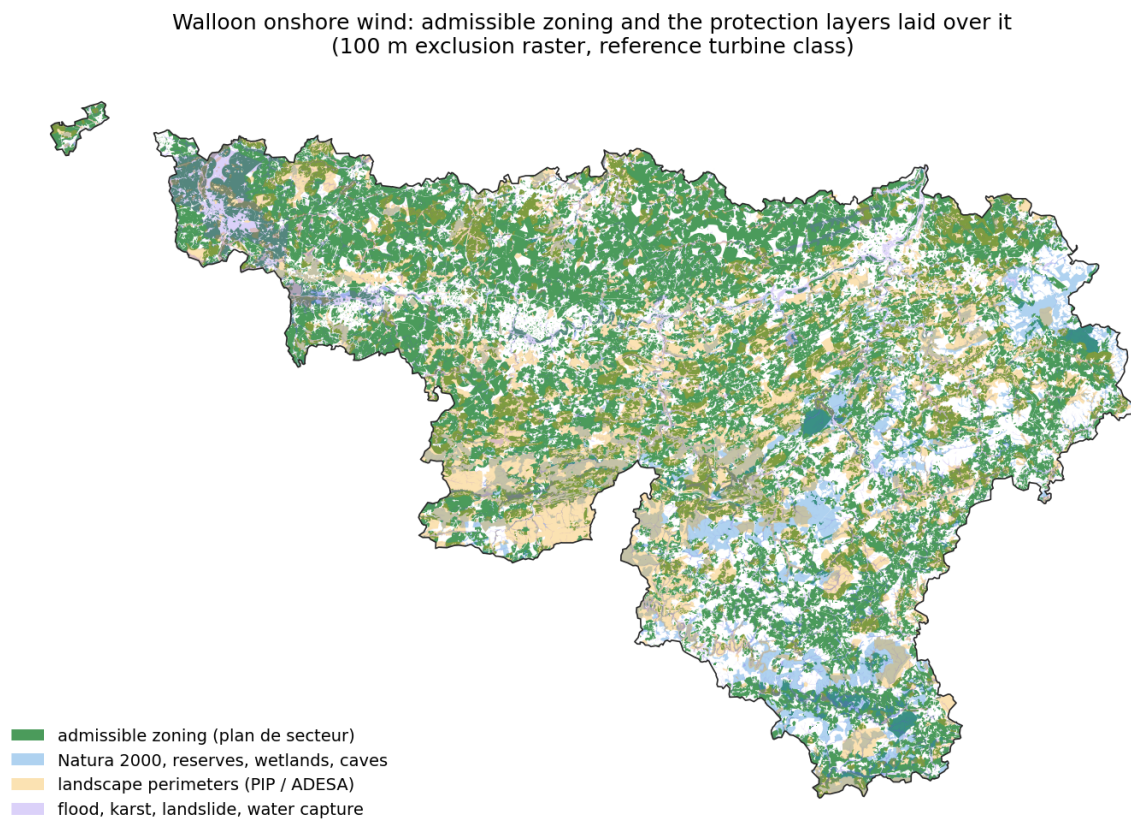


Figure 3: The admissible plan-de-secteur envelope with the protection layers laid over it, at the 100 m resolution of the exclusion raster.

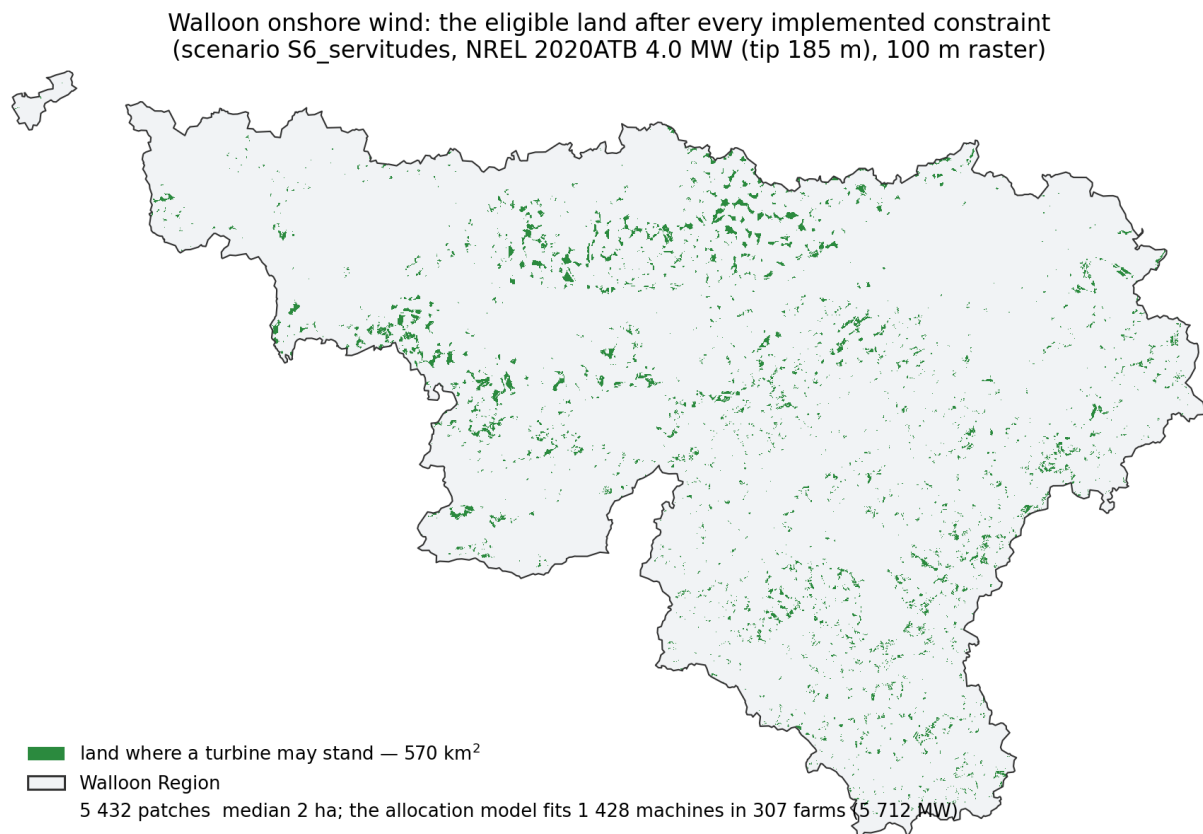


Figure 4: The answer, at 100 m resolution: the land that survives *every* implemented constraint. This is the raster the machines are placed on, and it is written out as `results/eligible_land_{turbine}.tif` so it can be opened in QGIS and checked parcel by parcel.

Table 7: Allocation results. *free* is the minimum-distance allocation alone; *grouped* keeps only machines in groups of 4 or more; *farm* allocates farm sites first; *farm + horizon* adds the 130° open-horizon criterion. Spacings as in Table 2.

turbine	model	spacing	d/D	variant	farms	machines	P [MW]
V112 / 3.0 MW (tip 136 m)	free	observed	4.3	row major	–	6 492	19 476
	grouped	observed	4.3	row major	–	6 377	19 131
	free	observed	4.3	random	–	5 523	16 569
	free	crosswind	5.0	row major	–	5 468	16 404
	grouped	crosswind	5.0	row major	–	5 345	16 035
	free	crosswind	5.0	random	–	4 639	13 917
	free	isotropic	5.9	row major	–	4 533	13 599
	grouped	isotropic	5.9	row major	–	4 404	13 212
	free	isotropic	5.9	random	–	3 912	11 736
	farm	observed	4.3	2.6 km apart	599	4 246	12 738
	farm + horizon	observed	4.3	2.6 km apart	410	3 070	9 210
	farm	observed	4.3	4.0 km apart	369	2 832	8 496
	farm + horizon	observed	4.3	4.0 km apart	313	2 436	7 308
	farm	observed	4.3	5.0 km apart	280	2 221	6 663
	farm + horizon	observed	4.3	5.0 km apart	258	2 068	6 204
	farm	observed	4.3	6.0 km apart	213	1 769	5 307
	farm + horizon	observed	4.3	6.0 km apart	208	1 733	5 199
	farm	crosswind	5.0	2.6 km apart	559	3 515	10 545
	farm + horizon	crosswind	5.0	2.6 km apart	394	2 590	7 770
	farm	crosswind	5.0	4.0 km apart	351	2 339	7 017
	farm + horizon	crosswind	5.0	4.0 km apart	293	1 979	5 937
	farm	crosswind	5.0	5.0 km apart	264	1 814	5 442
	farm + horizon	crosswind	5.0	5.0 km apart	238	1 670	5 010
	farm	crosswind	5.0	6.0 km apart	204	1 469	4 407
	farm + horizon	crosswind	5.0	6.0 km apart	199	1 439	4 317
	farm	isotropic	5.9	2.6 km apart	497	2 570	7 710
	farm + horizon	isotropic	5.9	2.6 km apart	365	1 910	5 730
	farm	isotropic	5.9	4.0 km apart	325	1 726	5 178
	farm + horizon	isotropic	5.9	4.0 km apart	277	1 482	4 446
	farm	isotropic	5.9	5.0 km apart	246	1 343	4 029
	farm + horizon	isotropic	5.9	5.0 km apart	228	1 244	3 732
	farm	isotropic	5.9	6.0 km apart	191	1 066	3 198
	farm + horizon	isotropic	5.9	6.0 km apart	187	1 045	3 135
NREL 2020ATB 4.0 MW (tip 185 m)	free	observed	4.3	row major	–	4 275	17 100
	grouped	observed	4.3	row major	–	4 127	16 508
	free	observed	4.3	random	–	3 668	14 672
	free	crosswind	5.0	row major	–	3 809	15 236
	grouped	crosswind	5.0	row major	–	3 659	14 636
	free	crosswind	5.0	random	–	3 251	13 004
	free	isotropic	5.9	row major	–	3 271	13 084
	grouped	isotropic	5.9	row major	–	3 121	12 484
	free	isotropic	5.9	random	–	2 818	11 272
	farm	observed	4.3	2.6 km apart	466	2 379	9 516
	farm + horizon	observed	4.3	2.6 km apart	333	1 753	7 012
	farm	observed	4.3	4.0 km apart	309	1 616	6 464
	farm + horizon	observed	4.3	4.0 km apart	263	1 396	5 584
	farm	observed	4.3	5.0 km apart	227	1 225	4 900
	farm + horizon	observed	4.3	5.0 km apart	218	1 182	4 728
	farm	observed	4.3	6.0 km apart	180	993	3 972
	farm + horizon	observed	4.3	6.0 km apart	176	970	3 880
	farm	crosswind	5.0	2.6 km apart	423	1 938	7 752
	farm + horizon	crosswind	5.0	2.6 km apart	307	1 428	5 712
	farm	crosswind	5.0	4.0 km apart	282	1 314	5 256
	farm + horizon	crosswind	5.0	4.0 km apart	245	1 149	4 596
	farm	crosswind	5.0	5.0 km apart	213	1 005	4 020
	farm + horizon	crosswind	5.0	5.0 km apart	204	968	3 872
	farm	crosswind	5.0	6.0 km apart	175	838	3 352
	farm + horizon	crosswind	5.0	6.0 km apart	171	820	3 280
	farm	isotropic	5.9	2.6 km apart	367	1 590	6 360
	farm + horizon	isotropic	5.9	2.6 km apart	272	1 190	4 760
	farm	isotropic	5.9	4.0 km apart	249	1 094	4 376
	farm + horizon	isotropic	5.9	4.0 km apart	218	963	3 852
	farm	isotropic	5.9	5.0 km apart	190	846	3 384
	farm + horizon	isotropic	5.9	5.0 km apart	179	804	3 216

Table 8: What each rule costs, reference turbine class and 5.0 D spacing. Only the first two rows are technical; the rest are landscape policy.

	farms	machines	P [MW]
free allocation, 750 m between machines	–	3 809	15 236
+ groups of 4 or more, farms 2.6 km apart	423	1 428	7 752
+ open horizon, 130° within 4 km (reference)	307	1 428	5 712
<i>policy sensitivity:</i> farms 4 km apart, with open horizon	245	1 149	4 596
<i>policy sensitivity:</i> farms 6 km apart, with open horizon	171	820	3 280

Partial constraints, 15 % (bracket 0–30 %). The method distinguishes integral from *partial* exclusions — medium-priority bird zones, bat-interest zones, migration corridors, the principal ecological structure, radar study zones between 8 and 16 km — and grants turbines falling in them a 25 % success rate. Their combined footprint is unknown; the central estimate assumes they cover a fifth of the surviving land, so the expected loss is $0.20 \times 0.75 = 15\%$.

Together they leave 76 % of the farm-model capacity in the central case. One further criterion — the azimuth of open horizon the framework asks to be preserved for each village — is a site-level test that neither an area method nor a farm-level allocation can represent; it is not given an allowance, and it pushes the same way as the other two.

6.5 The answer

Table 9: From eligible land to a technical potential, administrative Walloon Region, reference turbine class at 5.0 D spacing.

	P [MW]	
eligible land, all implemented constraints		561 km ²
× capacity density, 5.08 MW/km ²	2 849	<i>convention only</i>
free allocation, 750 m between machines	15 236	<i>land and wake bound</i>
+ grouping into farms	7 752	
+ open horizon, 130° within 4 km	5 712	<i>gross reference</i>
× residual allowance, central	4 370	central estimate
credible technical potential	3 199–5 712	
<i>if the 2013 indicative inter-distance were applied as a rule</i>	2 509–3 516	
PyPSA-Wal cap today	6 500	
standing fleet, end 2024	1 528	

Finding. The credible gross technical potential for onshore wind in the administrative Walloon Region is **3.2–5.7 GW**, central estimate 4.4 GW, on the reference turbine class and the 2024 regulatory framework. The range is the residual-allowance bracket of §6.4 applied to the reference case; it is not statistical. Applying the indicative 4–6 km inter-distance of the 2013 framework as a binding rule instead would give 2 509–3 516 MW, and choosing a machine spacing of 4.3 D or 5.9 D instead of 5.0 D would move the gross reference between 7 012 and 4 760 MW.

6.6 Full-load hours and energy

The capacity factor is computed on the 2013 ERA5/SARAH-3 cutout with the same layout weighting PyPSA-Eur uses. For the reference case it is 3 264 h, which with the farm-model capacity gives 18.64 TWh/yr gross and 14.26 TWh/yr after the residual allowance — against the 6.2 TWh/yr of the PNEC/PACE 2030 target.

Those full-load hours are high, and the reason is the machine, not the method. The reference class is an NREL 2020 ATB 4 MW turbine with a 150 m rotor — a specific power of 226 W m^{-2} , against 305 W m^{-2} for the V112 PyPSA-Eur uses by default — on a 110 m hub instead of 80 m. Low-specific-power machines on tall towers are exactly what raises full-load hours, and it is why the same workflow returns 2 470 h when the V112 is substituted into the same eligible land. The V112 number is the one that reconciles with §2.2 and with the model’s own resource.

Caveat. These are *gross* new-turbine full-load hours: no wake, availability or electrical losses are applied, which is PyPSA-Eur’s convention for `p_max_pu` and therefore the convention PyPSA-Wal needs. They are not comparable with the 1 885 h TIMES-WAL uses for the blended existing fleet, nor with what the standing Walloon fleet delivers; that discrepancy is analysed in `docs/renewable-potentials.md` §10. BREGILAB applies a calibrated performance ratio of 0.69 to the same kind of figure (§7.1), which is the right correction for a study reporting delivered energy.

Caveat. Only the 2013 weather year is available on this machine. The Walloon onshore-wind resource in 2013 is about 5% *better* than in 2010, the year the main model runs use (`docs/renewable-potentials.md` §10.1). Capacity is unaffected — it depends only on the land — but the energy column is optimistic by roughly that margin relative to a 2010 run.

Table 10: Reference scenario S6_servitudes across turbine classes and capacity densities. *derived* is $P_{\text{nom}}/(5D \times 7D)$; *pypsa eur* is PyPSA-Eur’s 3 MW km^{-2} ; *dense* is a $4D \times 6D$ upper bound. These are the area-times-density figures, retained for comparison with studies reported on that convention.

turbine	density	[MW km ⁻²]	area [km ²]	P [MW]	FLH [h]	E [TWh]
V112 / 3.0 MW (tip 136 m)	dense	8.00	611	4 890	2 470	12.1
	derived	6.83	611	4 176	2 470	10.3
	pypsa eur	3.00	611	1 834	2 470	4.5
NREL 2020ATB 4.0 MW (tip 185 m)	dense	8.00	561	4 487	3 264	14.6
	derived	5.08	561	2 849	3 264	9.3
	pypsa eur	3.00	561	1 683	3 264	5.5
NREL 2020ATB 5.5 MW (tip 208 m)	dense	8.00	534	4 273	3 337	14.3
	derived	5.13	534	2 740	3 337	9.1
	pypsa eur	3.00	534	1 602	3 337	5.3

Turbine class barely moves the energy, only the capacity. Across the three classes the energy yield per square kilometre of eligible land is 16.6–17.1 GWh/km², a spread of a few per cent, while the farm model on the same land returns 5 712–7 770 MW across the three classes. That is the expected physics: at fixed rotor-diameter spacing the swept area per square kilometre is a constant, so a low-specific-power machine trades megawatts for full-load hours at roughly constant energy. The choice of turbine class is therefore close to irrelevant for what the model needs from this study — annual energy — and decisive for the headline capacity number, which is a reason to report both.

7 Against the other estimates

7.1 BREGILAB / VITO Dynamic Energy Atlas

The other published bottom-up estimate of the Walloon onshore-wind potential is the BREGILAB project’s, computed with VITO’s Dynamic Energy Atlas (Clymans et al., 2022, WP3, Table 22). Its Table 22 gives Wallonia 1 010 MW current and 10 390 MW additional onshore wind; the 11.4 GW quoted in its running text is their sum, that is, a *gross* potential on the same convention

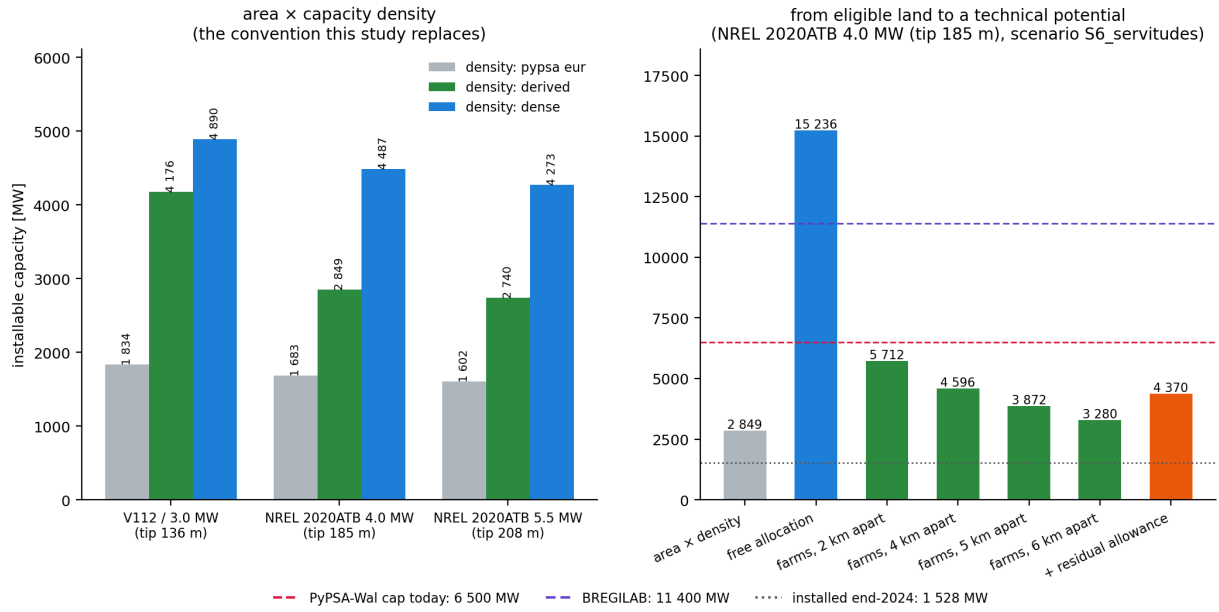


Figure 5: Installable capacity in the reference scenario, by turbine class and array spacing.

Table 11: This study against the figures currently used or published. All capacities are gross, i.e. they include the standing fleet.

source	basis	P [MW]	E [TW h]
PyPSA-Wal today (input_parameters_for_models.csv)	PNEC wallon / EDORA, expert judgement	6 500	21.22
PyPSA-Eur default land eligibility	this study, scenario S0, area x 3 MW/km ²	19 079	61.62
BREGILAB / VITO Dynamic Energy Atlas	Clymans et al. 2022, V112 at 5 D, gross	11 400	18.20
Walloon constraint set, area x density	this study, S6_servitutes, 5.08 MW/km ²	2 849	9.30
Walloon constraint set, free allocation	this study, S6_servitutes, 750 m between machines, no grouping rule	15 236	49.73
Walloon constraint set, farms allocated	this study, S6_servitutes, 3 km between farms, min. 4 machines	5 712	18.64
Walloon constraint set, machines placed, residual allowance	this study, central allowance for the families with no geometry	4 370	14.26
SPW/Gembloux favourable-zone update 2022, '180 m' case	site-by-site simulation, additional potential only	—	3.96
Installed fleet, end 2024	EDORA / Renouvelle	1 528	—
PNEC / PACE 2030 onshore-wind target	Walloon Government	—	6.20

as `p_nom_max`. It is $\times 0.75$ this study’s free allocation, $\times 2.0$ its farm allocation and $\times 2.6$ its central estimate, and the whole of the difference can be accounted for.

Method. BREGILAB delineates available space from the 2013 cadre de référence and the 2009 ICEDD study, on a 100 m grid, then runs an allocation algorithm that packs in as many Vestas V112 (3.3 MW, hub 94 m) as will fit subject to a minimum inter-turbine distance of $5D = 560$ m. Constraints are weighted rather than binary: air-traffic and military *orange* zones and the Walloon protected views around urban settlements carry a weight of 0.4 and bind only in combination with something else. Production is corrected by a performance ratio of 0.69, calibrated against Elia’s measured output — which is why its reported availability factor for the Walloon potential is 1 586 h and not the 2 298 h implied gross, a figure close to what this workflow gets for the same machine, and a useful independent check that the two resource models agree.

The bridge. Table 12 converts this study’s free allocation onto BREGILAB’s conventions. Both factors are exact.

Table 12: Bridge from this study’s free allocation to BREGILAB’s 11.4 GW. Each row applies one change; the last factor is the residual and is the constraint-set difference.

step	what changes	P [MW]	factor
this study, free allocation	NREL 2020ATB 4.0 MW (tip 185 m), 5 D between machines	15 236	
$\times$ turbine class	V112 instead: smaller setbacks, more eligible land	16 404	1.08
$\times$ constraint set	the residual: everything else	11 400	0.69
BREGILAB, Wallonia on-shore, gross	Clymans et al. (2022), BREGILAB WP3, VITO/EnergyVille, Table 22	11 400	0.75
this study, farm allocation + open horizon	the same land under the framework’s own landscape criterion	5 712	0.50

Turbine class, $\times 1.08$. The V112 is a 136 m machine. Under the 2024 framework its habitat setback is 568 m against 592 m for the reference class, so more land survives: 622 km² against 561, and at a fixed $5D$ the smaller rotor also packs tighter. The two effects compound, which is why the turbine class is not a detail when comparing two potentials. It is the only convention difference: both studies use a $5D$ spacing rule.

Constraint set, $\times 0.69$. What is left. It is *below* one: with BREGILAB’s machine and BREGILAB’s spacing, this study’s eligible land carries 16 404 MW against their 11 400 MW. Their constraint set is therefore the tighter of the two, which is the opposite of what the headline numbers suggest.

Why their constraint set is tighter, and ours is not uniformly looser. The components run in both directions and roughly cancel, with theirs ahead. BREGILAB applies civil-aviation and military *red* zones and SEVESO, pipeline and nuclear setbacks that this study does not have at all; it applies the 2013 habitat rule, which for its V112 on a 94 m hub is 600 m, against the 568 m the 2024 rule gives for the 80 m-hub V112 used here; and it applies a 2 km exclusion around nuclear facilities. Against that, it treats the Walloon landscape perimeters and protected views at weight 0.4 rather than as integral exclusions, applies no flood layer in Wallonia, and does not apply the 7 % slope criterion — which alone is worth 22.3 % of the land here.

Finding. The two studies do not disagree about Walloon geography. They disagree about what a technical potential is. BREGILAB reports the maximum number of machines that can be packed into the available space under a wake-avoidance distance; this study reports the number that can stand there once the framework’s own grouping and landscape rules are applied. On BREGILAB’s convention this study’s land gives 16 404 MW, more than their 11 400 MW; on this study’s convention the same land gives 5 712 MW. The factor between those two numbers — roughly three — is the whole of the disagreement, and it is a modelling convention, not a fact about Wallonia.

What was adopted from it. Three things, and together they changed the answer of this study more than any new data layer did:

1. **Allocating machines instead of multiplying area by a density.** This is BREGILAB’s method and it is better than an area calculation on fragmented land, and it removes the need for any patch-size screen.
2. **Graded rather than binary constraints.** BREGILAB’s weighting of consultation zones is the idea behind this study’s separate treatment of the aeronautical rings and behind the residual allowance for partial constraints.
3. **Validating against measured reality.** BREGILAB calibrates its production against Elia. This study cannot do that for a potential, but the same instinct produced the fleet calibration of §5, which removed two constraints that had no empirical support and confirmed the rest.

What was not adopted, and why. BREGILAB’s performance ratio of 0.69 is a calibration factor that absorbs both physical losses and the bias of its own wind model against Elia’s measurements. Applying it to a PyPSA-Eur profile would double-count, because PyPSA-Wal expects a gross `p_max_pu`. Its weighted-overlay treatment of soft constraints was also not reproduced literally: with only two soft families left unmapped here, an explicit allowance with a stated bracket (§6.4) is more transparent than a hidden weight.

7.2 The 2022 SPW/Gembloux favourable-zone update

That study reports 175 km² of *zones favorables strictes* and 272 km² of *zones favorables élargies* for its “180 m” case, and 3 959 GWh/yr of *additional* production from 457 potential turbines, cut down from 1 323 simulated ones by landscape inter-distance and encirclement criteria.

Three differences are structural:

- It applies the priority bird zones and the azimuth-of-open-horizon test that this study represents only by an allowance and by the inter-farm distance — the criteria responsible for its own cut from 1 323 to 457.
- It counts only *new* capacity, excluding the footprint of the standing fleet, whereas this study reports a gross potential.
- It applies a wind-resource threshold; this study applies none.

Adding the standing fleet back to its 457 turbines gives of order 4 GW on a comparable gross basis, against this study’s 4.4 GW central estimate, for the structural reasons listed above.

7.3 The model’s own cap

6 500 MW against this study’s central 4 370 MW is a factor of 1.5, and 6 500 MW lies inside the credible range 3 199–5 712 MW. Two qualifications matter.

First, the cap was constructed as a deployment expectation — what is plausible by 2030 plus the project pipeline — and this report produces a land-and-siting envelope. That the two land in the same place is a coincidence of construction, not a confirmation.

Second, 6 500 MW is above the gross reference case (5 712 MW) and is reachable only by dropping the open-horizon criterion altogether, which would give 7 752 MW, and then allowing

for none of the unmapped constraint families. If instead the 2013 indicative inter-distance were applied as a rule, the envelope would be 2 509–3 516 MW and the cap would be roughly twice it.

7.4 Three independent cross-checks

The conclusion of §6.5 rests on a chain — constraint set, eligible raster, machine allocation, farm allocation, residual allowance — and an error anywhere in it would move the answer. Three checks test the end of the chain against things that are known independently of it.

The size of a Walloon wind farm. The farm model predicts 4.7 machines per farm; the 652 standing turbines group into farms of 4.9 (§??). The model reproduces the layout the Walloon landscape and the Walloon rules actually produce, which is the single strongest evidence that the allocation is not an artefact. Note that this is not circular: the farm *radius* is calibrated on the fleet, but the number of machines that fit inside it is not — that comes from the eligible raster and the wake spacing.

The number of wind farms. The model places 307 farms against the 134 that exist, i.e. it says Wallonia could host of order twice as many wind farms as it has. That is the right order for a technical potential: much more than a factor two would be implausible in a Region this settled, and less than one would contradict the existence of a project pipeline.

The resulting deployment density, against a country that has built one. Germany ended 2024 with 63.55 GW of onshore wind in 28 717 machines on 357 600 km² — a deployed density of 178 kW/km² and 80.3 machines per 1 000 km². Wallonia today carries 90 kW/km², 0.51 times that. This study’s central estimate corresponds to 258 kW/km², 1.45 times Germany’s current deployed density, and the top of its range to 1.90 times.

Finding. A technical potential that lands at 1.45 times the density Germany has *already built* is in the right place: comfortably above what Wallonia has done, not far above what a comparable country has done, and well below the densities reached in Schleswig-Holstein or Lower Saxony. Had the method returned a figure at ten times Germany’s deployed density, it would have been describing an arithmetic exercise rather than a landscape.

Caveat. The farm allocation is greedy, not optimal: it is a heuristic that places farms in decreasing order of surrounding eligible land and never revisits a choice. An exact packing would fit somewhat more, so the farm figure is a slight under-estimate of what the rules permit — in the opposite direction to every other approximation in this report.

8 Hypotheses

Every choice below is a config entry, every one can be varied, and each is stated with the direction in which it moves the result.

Region. The administrative Walloon Region, not the model’s BEWAL node (§1.4). *A BEWAL figure would be about 10 % lower.*

Capacity comes from placing machines, not from multiplying area by a density (§2.4). *The density convention would give 2 849 MW, about half the reference case, for reasons that have nothing to do with the land.*

Minimum inter-turbine distance 5.0 $D = 750$ m, the nearest-neighbour distance of a $5D \times 7D$ array and the rule VITO uses. *The physical floor is about $3D$; the standing Walloon fleet sits at $4.3D$; the area-equivalent $\sqrt{5 \times 7} D$ would give 4 760 MW instead of 5 712 MW and is an over-reading (§2.4).*

Minimum farm size 4 machines, applied because 86 % of the standing fleet is in groups that size, not because the 2024 definition of a *parc éolien* forbids fewer — it does not, and it exempts machines above 3.2 MW in any case.

Farm radius and separation 1 300 m and 2.6 km, the observed p75 farm radius and twice it, which sits inside the observed p10–p25 band of farm separations (§??). *Calibrated, not imposed: the model forbids no farm geometry Wallonia has permitted.*

Landscape criterion the open horizon of the 2013 framework — 130° free within 4 km of every village above 10 ha. *The single most expensive siting rule, 26 % of the grouped capacity, and the one the standing fleet demonstrably respects (§??).*

The 2013 indicative 4–6 km inter-distance is *not* applied. It is an indicative reference subordinate to the impact assessment, it does not apply along motorways, and the 2024 framework that replaced the 2013 one does not carry it. *Applying it as a rule would give 2 509–3 516 MW instead of 4 370 MW.*

Habitat setback the 2024 framework, $500 + H/2$. *The 2013 rule ($4H$) is more restrictive for every class considered here.*

Scattered dwellings 400 m around every ICAR address point. The address layer includes non-residential buildings, so this is mildly conservative; the 2022 SPW study makes the same approximation.

Aeronautical servitudes the AOEM rings at 0 and 15 m only, after the fleet calibration of §5. *Including the 35 and 50 m rings would remove a further 89 km², against the evidence.*

Slope $\geq 7\%$ on the ERRUISSOL 10 m grid, majority rule over the 25 m samples in each 100 m cell. *The threshold is a 2013 criterion and is resolution-dependent; a coarser terrain model would exclude less.*

Radar and radio astronomy three circles at published coordinates with generic separation distances (§3.5). *Worth 8 km², so not load-bearing.*

Broad-leaved / coniferous split CORINE at 250 m, mixed stands counted as broad-leaved; CARTOFOR and WALOUS would both be better and neither is retrievable (§3.6). *Direction unknown.*

Residual allowance 10 % for priority bird zones and 15 % for partial constraints, both bracketed (§6.4). *Both reduce; the brackets are carried into the headline range.*

Azimuth of open horizon not represented. *Reduces; it is what cut the Region's own 2022 study from 1 323 turbines to 457.*

Runoff flood hazard excluded from the flood layer; the overflow extent at the extreme return period is used instead.

Geometry generalisation heavy polygon layers are requested with 25 m server-side generalisation — a quarter of the exclusion raster cell, so it cannot change the aggregate.

Existing wind farms are not excluded. The result is a *gross* potential: the standing 1 528 MW occupies land that is counted as available. This is the right convention for `p_nom_max`, which caps the total fleet, not additions.

Repowering is not a separate term. A land-and-siting study measures the envelope, not the turnover of the fleet inside it. EDORA's additional 3.8 GW by 2050 explicitly includes repowering, which is part of why that number and this one are not the same kind of quantity.

No grid-connection or economic screen. Distance to the HV network, connection cost and wind-speed profitability thresholds are not applied. *All reduce.*

Energy is gross. No wake, availability or electrical losses, matching PyPSA-Eur's `p_max_pu` convention. *A delivered-energy figure would be roughly 15 % lower; BREGILAB's calibrated performance ratio is 0.69.*

Weather year 2013. Capacity is unaffected; energy is about 5 % optimistic relative to the 2010 year the model runs use.

Finding. Almost every hypothesis in this list pushes the same way: down. The figure in §6.5 is an upper bound on a technical potential, and the constraint families that could not be mapped, the criteria that need a site-level model, and the economic screens that were not applied would all reduce it further.

9 What this means for `p_nom_max`

1. **The cap can be documented rather than cited.** 6 500 MW is reachable within this framework: it corresponds to the 4 km inter-farm distance with a small residual allowance, on 561 km² of eligible land carrying 245 farms of NREL 2020ATB 4.0 MW (tip 185 m). That is a statement the administration can contest on its merits. Replacing the CSV’s “PNEC wallon et EDORA” with that derivation costs nothing and removes the main weakness of the current number.
2. **If a single value is wanted, 4 370 MW is the defensible one** — the farm model at the framework’s permissive inter-distance with the central residual allowance. It is below the current cap by 1.5×, and the difference is entirely the two unmapped constraint families.
3. **Settle the region question before quoting anything externally.** A Region figure applied to the BEWAL node overstates the node by about 10 % (§1.4).
4. **Decide what `p_nom_max` is meant to be.** PyPSA-Wal already carries a build-rate limit and a 2030 corridor (`docs/renewable-potentials.md` §3–4) to represent deployment. If those are doing their job, `p_nom_max` should be the technical potential and the present value is the wrong kind of number in the right range. That is a modelling decision, not a GIS one.

10 What would sharpen the number further

None of these is a gap that invalidates the result; each would narrow the range.

1. **Ask SPW Énergie for the DEMNA priority-bird layer and the partial-constraint layers.** They exist and the favourable-zone cartography uses them. They would replace the residual allowance, which is the widest single term in the range, with geometry.
2. **Ask SPW Territoire how the 2024 framework’s encirclement test is applied in practice.** The 2013 text gave a 130° open horizon over 4 km and an encirclement examination over 9 km; the 2024 text asks for encirclement to be assessed without fixing either number. The open-horizon threshold is worth 26 % of the answer, and the 9 km examination is not represented at all.
3. **Request WALOUS 2018 as a grid, or CARTOFOR when it is published.** Either would replace a 2006 250 m raster with a Walloon product and would implement the CoDT broad-leaved rule where it currently is not applied at all (§3.6).
4. **Run the azimuth-of-open-horizon test on the placed farms.** The farm layout is written out as `results/placement_{turbine}.gpkg` precisely so that a landscape pass can be run on it. This is the one remaining criterion of the Walloon method that is not represented at all.
5. **Repeat on the 2010 weather year** once its cutout is available, to put the energy column on the same footing as the model runs.

A Full results

B Data provenance

All layers below were retrieved from the ESRI-REST services of the Géoportail de la Wallonie (<https://geoservices.wallonie.be>) under the CC-BY 4.0 licence.

Additional inputs: the slope-class grid `SOL_SOUS_SOL/ERRUISOL_CLASSE_PENTE`, decoded from the Region’s visualisation service (§3.5); the standing wind fleet from OpenStreetMap via the

Table 13: All scenarios and turbine classes at the reference capacity density, administrative Walloon Region.

scenario	turbine	area [km ²]	P [MW]	FLH [h]	E [TWh]
S0 pypsa eur	V112 / 3.0 MW (tip 136 m)	3 757	25 668	2 442	62.67
S1 zoning	V112 / 3.0 MW (tip 136 m)	8 521	58 223	2 464	143.44
S2 setbacks	V112 / 3.0 MW (tip 136 m)	1 511	10 325	2 454	25.34
S3 nature	V112 / 3.0 MW (tip 136 m)	1 316	8 990	2 460	22.11
S4 landscape	V112 / 3.0 MW (tip 136 m)	1 004	6 859	2 459	16.87
S5 risk	V112 / 3.0 MW (tip 136 m)	945	6 457	2 461	15.89
S6 servitudes	V112 / 3.0 MW (tip 136 m)	611	4 176	2 470	10.32
S0 pypsa eur	NREL 2020ATB 4.0 MW (tip 185 m)	3 757	19 079	3 230	61.62
S1 zoning	NREL 2020ATB 4.0 MW (tip 185 m)	8 521	43 277	3 257	140.96
S2 setbacks	NREL 2020ATB 4.0 MW (tip 185 m)	1 398	7 103	3 244	23.04
S3 nature	NREL 2020ATB 4.0 MW (tip 185 m)	1 211	6 152	3 251	20.00
S4 landscape	NREL 2020ATB 4.0 MW (tip 185 m)	923	4 689	3 251	15.25
S5 risk	NREL 2020ATB 4.0 MW (tip 185 m)	870	4 417	3 253	14.37
S6 servitudes	NREL 2020ATB 4.0 MW (tip 185 m)	561	2 849	3 264	9.30
S0 pypsa eur	NREL 2020ATB 5.5 MW (tip 208 m)	3 757	19 275	3 302	63.65
S1 zoning	NREL 2020ATB 5.5 MW (tip 208 m)	8 521	43 720	3 331	145.63
S2 setbacks	NREL 2020ATB 5.5 MW (tip 208 m)	1 341	6 879	3 317	22.82
S3 nature	NREL 2020ATB 5.5 MW (tip 208 m)	1 157	5 937	3 324	19.73
S4 landscape	NREL 2020ATB 5.5 MW (tip 208 m)	881	4 518	3 325	15.02
S5 risk	NREL 2020ATB 5.5 MW (tip 208 m)	830	4 257	3 326	14.16
S6 servitudes	NREL 2020ATB 5.5 MW (tip 208 m)	534	2 740	3 337	9.14

Overpass API (ODbL 1.0); ERA5/SARAH-3 `atlite` cutout for 2013 (Copernicus); CORINE Land Cover 2006 at 250 m, PyPSA-Eur data bundle; NUTS-3 shapes and the PyPSA-Wal onshore regions, PyPSA-Eur resources; turbine power curves from the `atlite` library (Vestas, NREL 2020 ATB).

Bibliographic reference for the comparison of §7.1: Clymans, W., Poelmans, L., Meyvis, L., et al. (2022), *BREGILAB WP3 RES Generation: Wind & PV deployment evolution and availability factor*, VITO/EnergyVille for the FPS Economy.

Table 14: Source datasets, as retrieved.

dataset	service / layer	features	retrieved
Points d'adresse ICAR	DONNEES_BASE/ICAR_- ADR_PT/1	1 512 246	2026-09-22
Périmètres d'intérêt paysager de l'inventaire ADESA	AMENAGEMENT_TERRI- TOIRE/ADESA/1	1 142	2026-09-22
Carte d'évaluation pour les obstacles aéronautiques (DGTA)	AMENAGEMENT_TERRI- TOIRE/AOEM/0	6 435	2026-09-22
Cavités souterraines d'intérêt scientifique	FAUNE_- FLORE/CONSNAT/0	80	2026-09-22
Zones inondables — débordement, période de retour extrême	EAU/ZI/2	39 893	2026-09-22
Biens classés — ensembles architecturaux	AMENAGEMENT_TERRI- TOIRE/BC_PAT/2	53	2026-09-22
Biens classés — zones de protection	AMENAGEMENT_TERRI- TOIRE/BC_PAT/4	367	2026-09-22
Biens classés — sites classés	AMENAGEMENT_TERRI- TOIRE/BC_PAT/1	1 259	2026-09-22
Patrimoine mondial UNESCO — zones tampon	AMENAGEMENT_TER- RITOIRE/PAT_MND_UN- ESCO/1	24	2026-09-22
Zones de contraintes karstiques	AMENAGEMENT_TERRI- TOIRE/CONTR_KARST/0	449	2026-09-22
Contraintes physiques liées aux éboulements	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/0	997	2026-09-22
Réseau Natura 2000 — périmètres des sites en vigueur	FAUNE_- FLORE/NATURA2000/9	240	2026-09-22
Plan de secteur — lignes électriques haute tension	AMENAGEMENT_TERRI- TOIRE/PDS/12	1 016	2026-09-22
Plan de secteur — périmètres d'intérêt paysager	AMENAGEMENT_TERRI- TOIRE/PDS/17	1 456	2026-09-22
Plan de secteur — réseau ferroviaire	AMENAGEMENT_TERRI- TOIRE/PDS/11	911	2026-09-22
Plan de secteur — réseau routier	AMENAGEMENT_TERRI- TOIRE/PDS/9	5 420	2026-09-22
Plan de secteur — voies navigables	AMENAGEMENT_TERRI- TOIRE/PDS/13	1	2026-09-22
Plan de secteur — zones d'affectation	AMENAGEMENT_TERRI- TOIRE/PDS/22	43 797	2026-09-22
Réserves naturelles agréées	FAUNE_- FLORE/CONSNAT/3	264	2026-09-22
Réserves naturelles domaniales	FAUNE_- FLORE/CONSNAT/2	306	2026-09-22
Réserves forestières	FAUNE_- FLORE/CONSNAT/1	20	2026-09-22
Versants supérieurs à 30 degrés	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/1	2 834	2026-09-22
Zones de prévention rapprochée (IIa) des captages	EAU/PROTECT_CAPT/2	736	2026-09-22
Zones humides d'intérêt biologique	FAUNE_- FLORE/CONSNAT/4	73	2026-09-22